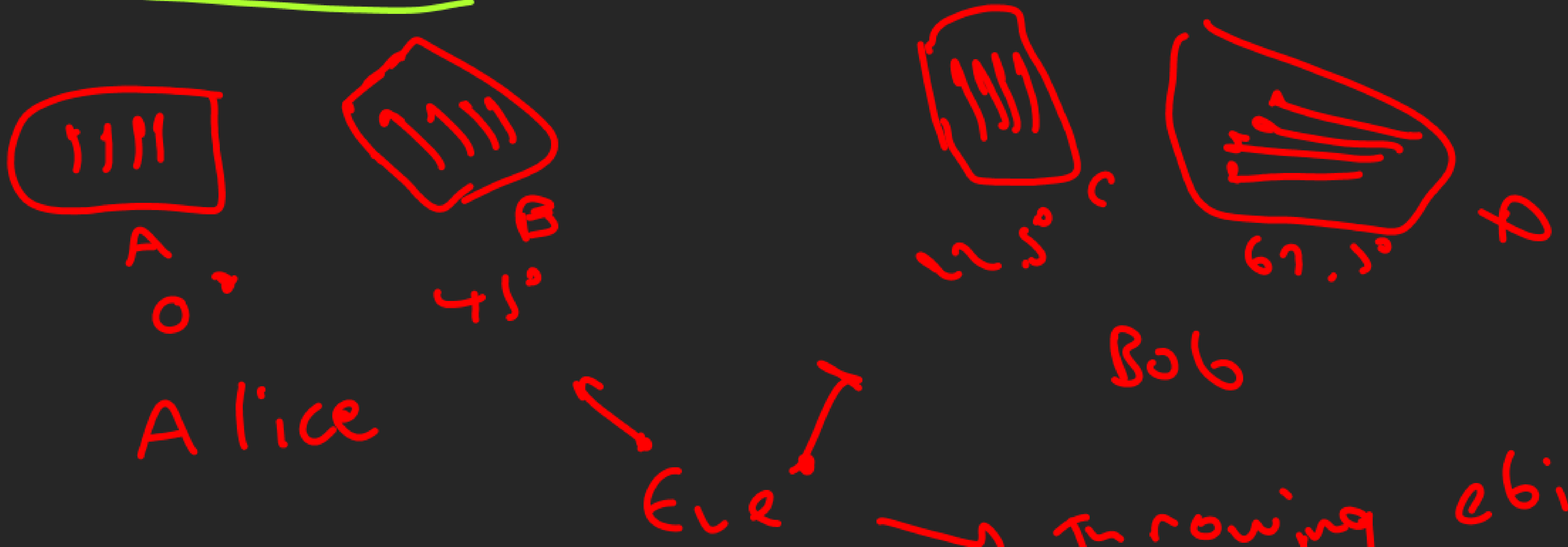
Bell Inequality



→ Polarisers

→ Throwing ebits

measured ? Spin up → 1

Spin down → −1

$$Q = AC + BC + BD - AD$$

For $A, B, C, D = \pm 1 \Rightarrow Q = \pm 2$

Bell's Inequality

$$\langle AC\rangle + \langle BC\rangle + \langle BD\rangle - \langle AD\rangle \leq 2$$

Experimentally ~ $\langle AC\rangle + \langle BC\rangle + \langle BD\rangle - \langle AD\rangle = 2.8 \approx 2\sqrt{2}$

We'll use the observables $\rightarrow A = Z$

$\rightarrow B = X$ $\rightarrow$ Eigenvalues $\downarrow$ ± 1

$\rightarrow C = \dfrac{X+Z}{\sqrt{2}}$

$\rightarrow D = \dfrac{X-Z}{\sqrt{2}}$

$z \to 0°$ $\quad x \to 90°$ $\quad \frac{x+z}{\sqrt{2}} \to 45°$ $\quad \frac{x-z}{\sqrt{2}} \to 135°$

↳ physical angles are half of the angles on Bloch sphere

Avg. of Outcomes $= \sum_m p(m)\cdot m = \sum_m \langle\psi|M_m^\dagger M_m|\psi\rangle\, m$

→ $\langle M\rangle$

$= \langle\psi|\left(\sum_m m\, M_m^\dagger M_m\right)|\psi\rangle$

$$M = \sum_m m\, M_m^\dagger M_m$$

$$= \sum_\lambda \lambda\, |\lambda\rangle\langle\lambda|$$

Eigenvalues

Eigenvectors

$\langle M\rangle = \langle\psi|M|\psi\rangle$

↳ Special Theorem of Linear Algebra

$$\langle AC\rangle = \langle Z \otimes \left(\frac{X+Z}{\sqrt{2}}\right)\rangle = \frac{\langle Z \otimes X\rangle}{\sqrt{2}} + \frac{\langle Z \otimes Z\rangle}{\sqrt{2}}$$

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad \therefore \frac{\langle\psi|(Z \otimes X)|\psi\rangle}{\sqrt{2}} + \frac{\langle\psi|(Z \otimes Z)|\psi\rangle}{\sqrt{2}}$$

$$\boxed{(A \otimes C)(B \otimes D) = AB \otimes CD}$$

$\langle Z \otimes X\rangle = 0 \rightarrow$ Bloch Sphere

$\langle Z \otimes Z\rangle = 1$

$\cos(90^\circ) = 0$

$$\langle AC\rangle = \frac{1}{\sqrt{2}} \quad \langle BC\rangle = \frac{1}{\sqrt{2}} \quad \langle BD\rangle = \frac{1}{\sqrt{2}} \quad \langle AD\rangle = -\frac{1}{\sqrt{2}}$$

$= \langle AC \rangle + \langle BC \rangle + \langle BD \rangle - \langle AD \rangle = 2\sqrt{2}$

Bloch sphere

$\langle AC \rangle = \cos 45^\circ = \frac{1}{\sqrt{2}}$